VSEC Commands and Files

Assignment No.4(Jenkins)

If Jenkins is not started in PC then use this:

```
sudo systemctl start jenkins.service
sudo systemctl status jenkins.service
```

Assignment No.5(Docker)

Dockerfile:

```
FROM python:3.9
WORKDIR /app
COPY requirements.txt .
RUN pip install -r requirements.txt
COPY app.py .
CMD ["python", "app.py"]
```

Open terminal inside your project folder:

```
docker build -t my-ai-app .
```

Run Docker Container

```
docker run -p 5000:5000 my-ai-app
```

Assignment No.6(Docker compose)

Check version:

```
docker-compose --version
```

Docker compose install(if not shown):

```
sudo apt install docker-compose -y
```

docker-compose.yml

```
version: "3"
services:

  mongodb:
    image: mongo
    container_name: mongo_db
    ports:
      - "27018:27017"

  model:
    build: ./model
    ports:
      - "5001:5000"
    depends_on:
      - mongodb

  frontend:
    build: ./frontend
    ports:
      - "8081:80"
    depends_on:
      - model
```

Dockerfile (frontend)

```
FROM nginx:alpine
COPY index.html /usr/share/nginx/html/index.html
```

Build and Run Containers

docker-compose up –build

Verify Running Containers

sudo docker ps

Test MongoDB

Access MongoDB container:

sudo docker exec -it mongo_db mongosh

Stopping Containers

docker-compose down

Assignment No.7(minikube)

Start Cluster (Minikube)

minikube start

Open Kubernetes Dashboard (UI)

minikube dashboard

Build Image (1 Command Required)

eval $(minikube docker-env)
docker build -t ai-model .

Upload YAML

apiVersion: apps/v1

kind: Deployment

```
metadata:
  name: ai-model-deployment
spec:
  replicas: 2
  selector:
    matchLabels:
      app: ai-model
  template:
    metadata:
      labels:
        app: ai-model
    spec:
      containers:
      - name: ai-container
        image: ai-model
        imagePullPolicy: Never
        ports:
        - containerPort: 5000
```

Create Service

```
apiVersion: v1
kind: Service
metadata:
  name: ai-model-service
spec:
  type: NodePort
  selector:
    app: ai-model
  ports:
```

```
  - port: 5000
    targetPort: 5000
    nodePort: 30007
```

Assignment No.8(minikube hpa)

Start Cluster (Minikube)

minikube start

Open Kubernetes Dashboard (UI)

minikube dashboard

Build Image (1 Command Required)

```
eval $(minikube docker-env)
docker build -t ai-model .
minikube image load ai-model
```

Using YAML:

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: ai-app
spec:
  replicas: 2
  selector:
    matchLabels:
      app: ai-app
  template:
    metadata:
```

```
    labels:
      app: ai-app
  spec:
    containers:
    - name: ai-container
      image: ai-model
      ports:
      - containerPort: 5000
```

Create Service

```
apiVersion: v1
kind: Service
metadata:
  name: ai-service
spec:
  type: NodePort
  selector:
    app: ai-app
  ports:
    - port: 5000
      targetPort: 5000
      nodePort: 30007
```

Access Application

Get URL:

minikube service ai-service

Enable Metrics Server

minikube addons enable metrics-server

Verify: (wait 5 min)

kubectl top pods

Create Horizontal Pod Autoscaler

kubectl autoscale deployment ai-app --cpu-percent=50 --min=2 –max=10

OR

Create HPA

```
apiVersion: autoscaling/v2
kind: HorizontalPodAutoscaler
metadata:
  name: ai-app-hpa
spec:
  scaleTargetRef:
    apiVersion: apps/v1
    kind: Deployment
    name: ai-app
  minReplicas: 2
  maxReplicas: 10
  metrics:
  - type: Resource
    resource:
      name: cpu
      target:
        type: Utilization
        averageUtilization: 50
```

Monitor Autoscaling

Command:

kubectl get hpa

Generate Load on system:

```
for i in {1..200}; do curl http://$(minikube ip):30007; done
```

OR

```
while true; do
  curl -s http://$(minikube ip):30007 > /dev/null
  sleep 1
done
```

Check Scaling (Pods Increase)

Command:

```
kubectl get pods -w
```

Check the changes after the load:

```
kubectl describe hpa  <service-name>
```

Check Logs & Health

```
kubectl describe pod <pod-name>
```

Cluster Monitoring

```
kubectl top nodes
```

Assignment No.9(Jenkins)

Check path:

```
pwd
```

Build Triggers

✓ **Enable:**

Poll SCM

Schedule:

H/1 * * * *

Allow permission for directory:

sudo chown -R jenkins:jenkins <path>